

BIOS vs UEFI

Enterprise Boot Firmware Comparison

ITEP 414 – System Administration and Maintenance | Week 3

Prepared by: Ralph Rio H. Bautista

1. Introduction

BIOS and UEFI are firmware environments that initialize hardware and begin the operating-system boot process. BIOS is the traditional firmware model, while UEFI is the modern replacement used by most current computers.

2. Comparison

Feature	BIOS	UEFI
Definition	Legacy firmware interface	Modern firmware interface/specification
Boot Process	Legacy boot code from the boot device.	Loads an EFI boot application
Disk Support	Traditional MBR commonly limits bootable disks to about 2 TB.	GPT supports very large disks.
Partition Style	Typically MBR.	Typically GPT.
Security	Limited built-in boot security.	Supports Secure Boot.
Speed	Older initialization model.	Often faster/more flexible on modern systems
Advantages	Mature and compatible with legacy systems	Modern architecture, GPT, Secure Boot.
Disadvantages	Legacy limitations and fewer modern features	More complex and dependent on modern firmware/OS support.
Modern Usage	Mostly legacy compatibility.	Standard on modern PCs and servers.

3. Why UEFI Replaced BIOS

UEFI has largely replaced traditional BIOS because modern systems need larger storage support, improved boot flexibility, and stronger platform security. BIOS

commonly works with MBR, while UEFI systems typically use GPT. GPT supports many more partitions and avoids the traditional MBR 2 TB boot-disk limitation.

UEFI also provides a standardized environment capable of loading boot applications and supporting Secure Boot. Secure Boot helps ensure that only trusted boot software is executed during startup.

4. System Administration Impact

Administrators need to know whether a machine uses legacy BIOS or UEFI because firmware mode affects partitioning, boot loaders, recovery procedures, and operating-system installation. A mismatch can result in boot failures.

5. Key Takeaways

- BIOS is the legacy firmware model.
- UEFI is the modern firmware standard.
- BIOS commonly uses MBR; UEFI commonly uses GPT.
- UEFI supports Secure Boot
- Firmware knowledge is important for OS installation and troubleshooting.

References

UEFI Forum – <https://uefi.org/>

Microsoft Learn – <https://learn.microsoft.com/>

Ubuntu Documentation – <https://documentation.ubuntu.com/>